

Project ID :

1. Topic (12 words max)

Investor Profiling and Recommendation System for International Capital Markets

2. Research group the project belongs to

TIM - Technology Integration and Management

3. Specialization of the project belongs to

Information System Engineering(ISE)

4. If a continuation of a previous project:

Project ID	-
Year	-

5. Brief description of the research problem including references (200 – 500 words max) – references not included in word count.

The global investment landscape is growing in complexity, with investors seeking opportunities across asset classes, sectors, and jurisdictions. Advisory firms are faced with a range of issues regarding meeting these demands:

Volume of options: Thousands of international equities, commodities, indices, and funds are available, and manual selection and comparison are no longer feasible [1].

Inconsistencies in profiling: Traditional investor profiling relies on long questionnaires and advisor judgment. These are prone to subjectivity and are time-consuming [2], [3].

Cross-border complexity: International investments involve various currencies, regulatory settings, and market norms. Small and medium advisory organizations may lack resources to handle this effectively [4].

Responsible investment themes: Investors more and more demand Environmental Social Governance (ESG) alignment. The majority of tools do not integrate these variables in an organized format [5].

Existing robo-advisory and web-based platforms focus on thin local markets or require costly, coding-dense development. There is therefore space for affordable, adaptable solutions that can be rolled out by smaller advisory teams [6].

The above-stated research problem was thus identified: How can a low-code/no-code system be created to provide personalized, explainable, and globally inclusive investment advice and remain accessible to smaller advisory firms?

This project will implement and launch a system of this kind, demonstrating the application of low-code/no-code technology to financial technology. It will integrate an investor-profiling front end, a filtered dataset of investments, and a rule-based recommendation engine. The study will validate the system through simulated investor profiles and minimal usability feedback, closing the gap between advanced advisory need and access to low-code technologies.

References

[1]	World Bank, ""Global Financial Development Report 2023,"" World Bank Publications, 2023.
[2]	H. Markowitz, ""Portfolio Selection,"" <i>The Journal of Finance</i> , vol. 7, no. 01, p. 77–91, 1952.
[3]	CFA Institute, Standards of Practice Handbook, 2020.
[4]	International Monetary Fund (IMF), "World Economic Outlook: Navigating Global Divergences," 2022.
[5]	MSCI, "ESG Ratings Methodology," 2021.
[6]	PwC, "Asset and Wealth Management Revolution 2020," PwC Global, 2020.

6. Brief description of the nature of the solution including a conceptual diagram (250 words max)

The solution proposed is a low-code/no-code tool employed to operationalize investor profiling and recommendation in a transparent and simple way. It is not simply a technology artefact but also a proof of concept on how digital platforms can transform financial advisory procedures.

It has four modules that are integrated into it:

Investor Profiling Module: identifies key characteristics such as risk tolerance, investment time horizon, preferred industries or geography, need for sustainability (ESG), and rough investment capacity. The plan will balance completeness with ease of use to give rapid profiles without lengthy questionnaires.

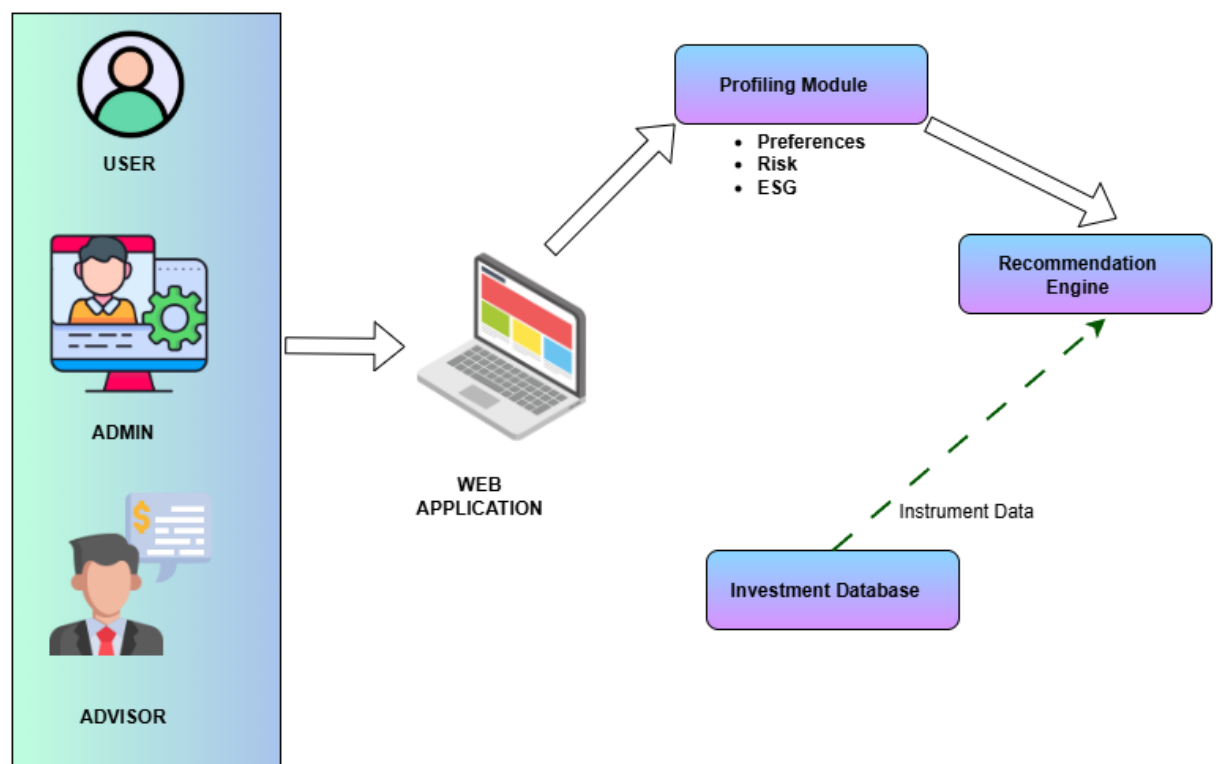
Investment Database: is a highly selective database of approximately 400 instruments (indices, funds/ETFs, commodities, stocks). Each entry holds crucial financial and

context data including indicative risk level, synthetic ESG score, currency, industry, instrument type, country/region, name, ticker, and minimum investment size.

Recommendation Engine: uses a rule-based weighted score system to compare investor profiles against products on offer. It provides ranked answers with short reasons to maintain transparency and build user trust.

User Dashboard and Reporting: enables investors to see their matches in diagrammatic form, compare alternative choices, mark favorites, and export results in summary reports for future reference or for discussion with advisors.

A low-code/no-code environment (e.g. Microsoft Power Apps) will be used as it is selected due to its quick developmental and integration capabilities. This methodology emphasizes the role of advisory firms (in particular, smaller firms) to implement functional decision-support systems that do not require significant programming resources to reduce costs and expand access.



7. Brief description of specialized domain expertise, knowledge, and data requirements
(300 words max)

This project involves the merging of knowledge in the financial markets, information systems, and implementation of technology to achieve both technical and user relevance.

To structure investor tolerance, horizon and ESG preferences, the Investor Profiling Module presupposes the understanding of behavioral finance and risk assessment frameworks. Survey and user-interface design is also necessary to ensure the capture of inputs in a tight, but complete manner.

Investment Database component will rely on the knowledge of the international capital markets, namely, equities, commodities, indices, and funds. It demands categorizing tools using the characteristics like ticker, industry, region, currency, indicative risk level, and ESG score. It needs data cleaning and curation as well as schema design to generate an identical dataset to be integrated into the system.

Decision-support logic has to be used in the **Recommendation Engine**. Though not machine learning but rule-based it requires the knowledge of scoring models, weighting schemes, and principles of transparency so that the results could be explained. The concept of responsible investment (e.g., ESG integration) knowledge is needed in order to make the recommendations adaptable to contemporary advisory situations.

The start of the **Dashboard and Reporting Interface** involves realistic competence in the low-code / no-code application of Microsoft Power Apps Information visualization knowledge and usability assessment would mean that the recommendations are presented in a logical and coherent manner and understandable by end users.

The knowledge requirements are:

- Understanding of risk profiling procedures, international financial instruments, and ESG investing.
- Proficiency in rule-based system design, data modelling, and data preparation.
- Languages Low-code/no-code competence, interface and workflow development.

Data requirements include:

- A about 400-strong simulated investing portfolio that includes realistic stocks, commodities, index funds, and fund/ETFs.
- More open-source data that has been verified and enhanced, such as that from the World Bank, IMF, and Yahoo Finance.
- Artificial scoring of the ESG and risk values; there is a prospective capability of embedding licensed information.

8. Objectives and Novelty

Main Objective To design and implement low-code/ no-code investor profiling and recommendation system to be applied in international capital markets, and to measure its performance.			
Member Name with Registration No	Sub Objective	Tasks	Novelty
Dahanayake K.A IT21232404	Construct and validate dataset of at least 300 instruments	- Collect, clean, and structure dataset of stocks, commodities, indices, and funds. - Set up attributes such as ticker, industry, region, currency, risk, and ESG. - Make consistent and system-integration ready.	Curated multi-asset, international dataset with ESG and risk scoring; structured for integration
	Design system architecture and profiling model	- Investor profiling form recording risk, horizon, ESG, industry, and region preferences. - Organize investor inputs into dataset attributes. - Define matching rules and score logic.	Transparent profiling logic that is simple but tailored, yet capable of handling various types of investors.

9. Individual component description of how it is complied with the specialization.

Member Name with Registration No	Description
Dahanayake K.A IT21232404	Implementing information systems principles, data modelling, and low-code/no-code application development. Essential components of technology integration and management is necessary to create the Investor Profiling and Recommendation System. A multi-asset international dataset must be curated and organised, profiling methods must be designed in accordance with risk assessment and ESG preferences, and a rule-based recommendation engine must be put into place. This illustrates how corporate information systems expertise can be combined with useful technological platforms to produce a useful financial decision-support tool.

10. Supervisor details

	Title	First Name	Last Name	Signature
Supervisor	Ms.	Jenny	Krishara	<i>Jenny</i> 2025/09/00
Co-Supervisor				
External Supervisor				
Summary of external supervisor's (if any) experience and expertise				

This part is to be filled by the Topic Screening Staff members.

- a) Does the chosen research topic possess a comprehensive scope suitable for a final-year project?

Yes	☒	No	☐
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- b) Does the proposed topic exhibit novelty?

Yes	☒	No	☐
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- c) Do you believe they have the capability to successfully execute the proposed project?

Yes	☒	No	☐
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- d) Do the proposed sub-objectives reflect the students' areas of specialization?

Yes	☒	No	☐
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- e) Supervisor's Evaluation and Recommendation for the Research topic:

Need to gather the proper dataset
for the research.

Acceptable: Mark/Select as necessary

Topic Assessment Accepted	✓
Topic Assessment Accepted with minor changes*	
Topic Assessment to be Resubmitted with major changes*	
Topic Assessment Rejected. Topic must be changed	

* Detailed comments given below

Comments

Need to identify the approach and the data required for the research.

Staff Member's Name	Signature
Ms. Jenny Krishara	Jenny 2025/09/26

*Important:

1. According to the comments given by the evaluator, make the necessary modifications and get the approval by the **Evaluator**.
2. If the project topic is rejected, identify a new topic, and request the RP Team for a new topic assessment.